

WASP-104b

R-2471

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_180402 · created 2026-08-23T23:22:32+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458210.6669 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.039
Transit depth (Rp/Rs)^2	2.25 +/- 1.18 [%]
Orbital Inclination (inc)	81.54 +/- 2.41
Airmass coefficient 1 (a1)	0.85 +/- 0.014
Airmass coefficient 2 (a2)	0.138 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.65 %
Best Comparison Star	#1 - [595, 42]
Optimal Aperture	3.24
Optimal Annulus	11.79
Transit Duration (day)	0.047 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.039 vs 0.1214 (deviation 0.73σ)
Duration fit vs published	0.047 d vs 0.0742 d (deviation 0.85σ)
Mid-time O–C	-3.0 min
Depth SNR	3.8
Residual scatter	1.65 %

Light curve

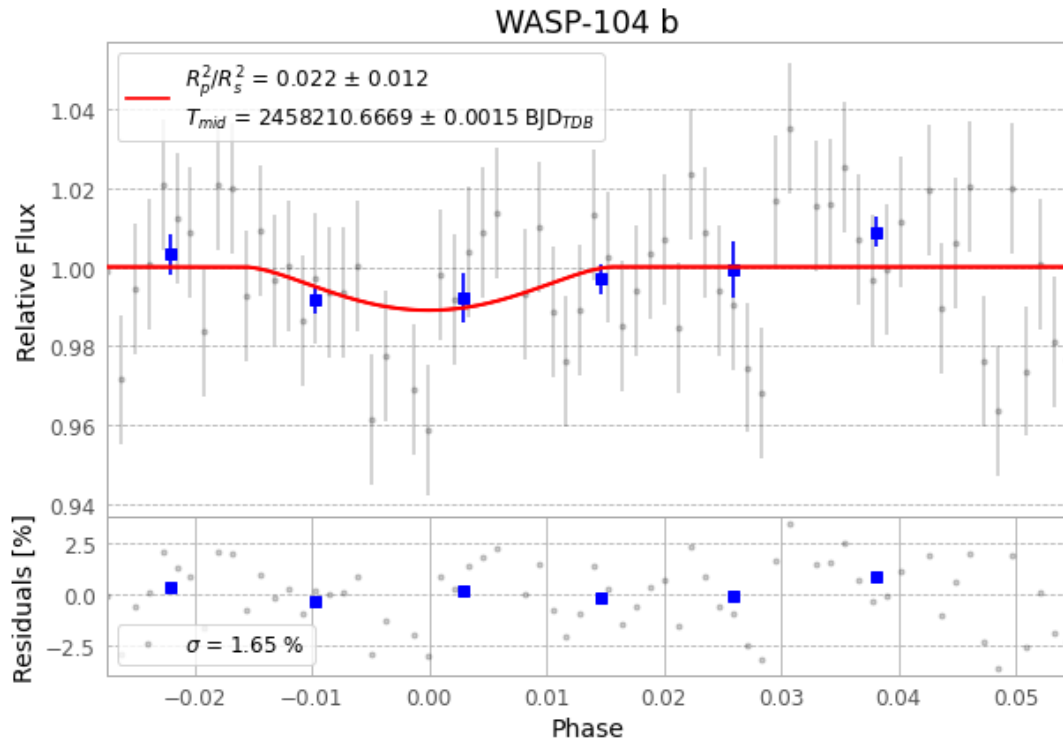


Figure 1 — FinalLightCurve_WASP-104 b_2018-04-01.png (SHA-256 daed96258dcf36d1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 190], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8c5af905948ab0ed...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

70 FITS frames (46 MB), WASP-104180402024212.FITS → WASP-104180402060912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-180402.log (62e55fceb41...), FinalLightCurve_WASP-104 b_2018-04-01.png (daed96258dcf...), FinalLightCurve_WASP-104 b_2018-04-01.csv (02e2a56b92b4...)

Compute resources

Wall time 133.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 23:22Z.